ExpensePro – MERN Stack Expense Management System

Project Title: ExpensePro
Developed By: Kamal Sagar
Technology Stack: MERN Stack (MongoDB, Express.js, React.js, Node.js)
Authentication: JWT Access & Refresh Token
Project Type: Full Stack Web Application

ExpensePro — Personal Finance Management System

In modern financial management, individuals need efficient tools to track their income, expenses, and budgets. Many people struggle to monitor their spending habits, which often leads to poor financial planning and overspending.

ExpensePro is a full-stack web application developed using the **MERN stack** to help users manage their personal finances effectively. The system allows users to track expenses, manage income, create budgets, and analyze spending patterns through visual dashboards.

The platform provides a modern user interface and secure backend architecture to ensure efficient and safe financial management. With features such as **JWT authentication, real-time expense tracking, budget management, and spending insights**, ExpensePro provides a complete solution for personal finance management.

Problem Statement

Many individuals do not maintain proper records of their income and daily expenses. As a result, they face difficulties understanding their financial habits and planning their monthly budgets.

Traditional methods such as spreadsheets or manual record-keeping are time-consuming and inefficient. Additionally, these methods lack features like real-time insights, category tracking, and automated budget monitoring.

To address these issues, there is a need for a digital platform that enables users to:

- Track their expenses easily
- Monitor their income
- Manage monthly budgets
- Visualize financial data

The Solution

ExpensePro solves these challenges by providing an intuitive and secure financial tracking platform.

Objectives of the Project

The main objective of the ExpensePro project is to develop a secure and efficient expense management system using modern web technologies.

Key objectives include:

1. **Full-Stack System**
   To create a full-stack expense tracking system using the MERN stack.
2. **Secure Authentication**
   To implement secure user authentication using JWT tokens.
3. **Income & Expense Tracking**
   To allow users to track income and expenses efficiently.
4. **Category Tracking**
   To provide category-based expense tracking.
5. **Budget Monitoring**
   To enable users to set and monitor monthly budgets.
6. **Financial Insights**
   To display financial insights through charts and analytics.
7. **Secure API Communication**
   To implement secure API communication between frontend and backend.

The project aims to help users improve financial awareness and budgeting habits.

Technology Stack — Frontend

ExpensePro is developed using the **MERN stack**, which provides a powerful ecosystem for building modern web applications.

Frontend Technologies

React.js

React.js is used to build dynamic and responsive user interfaces. It enables component-based architecture and improves performance through virtual DOM rendering.

Tailwind CSS

Tailwind CSS is used for modern UI design and responsive layout styling.

JavaScript

JavaScript provides interactive features and handles frontend logic.

Technology Stack — Backend

Backend Technologies

Node.js

Node.js provides the runtime environment for executing JavaScript on the server side.

Express.js

Express.js is used to build REST APIs and manage server routes.

Technology Stack — Database

Database

MongoDB

MongoDB is a NoSQL database used to store user data, expenses, budgets, and financial records.

System Architecture

ExpensePro follows a **three-tier architecture** consisting of:

Client Layer (Frontend)

The client layer is developed using React.js and is responsible for displaying the user interface.

Application Layer (Backend APIs)

The backend server built with Node.js and Express handles API requests, authentication, and business logic.

Database Layer (MongoDB)

MongoDB stores application data including user accounts, expenses, income records, and budget information.

Communication between frontend and backend occurs through **REST APIs**.

Authentication & Security

Security is a key component of the ExpensePro system. The application uses **JWT (JSON Web Token)** authentication to secure API requests.

Authentication Flow

01

User registers or logs in.

02

Server generates **Access Token** and **Refresh Token**.

03

Access Token is used to access protected APIs.

04

Refresh Token generates a new access token when it expires.

Security Features

- JWT Access Token authentication
- Refresh token system
- Token expiration handling
- Secure API endpoints
- Rate limiting to prevent abuse

Environment variables used include:

```
ACCESS_TOKEN_EXPIRE
REFRESH_TOKEN_EXPIRE
```

These measures ensure safe user authentication and secure communication.

Core Features of ExpensePro

ExpensePro provides several powerful features for financial management.

User Authentication

Users can securely register and log in to the system.

Expense Tracking

Users can add, update, and delete expenses.

Income Tracking

Users can record their monthly income.

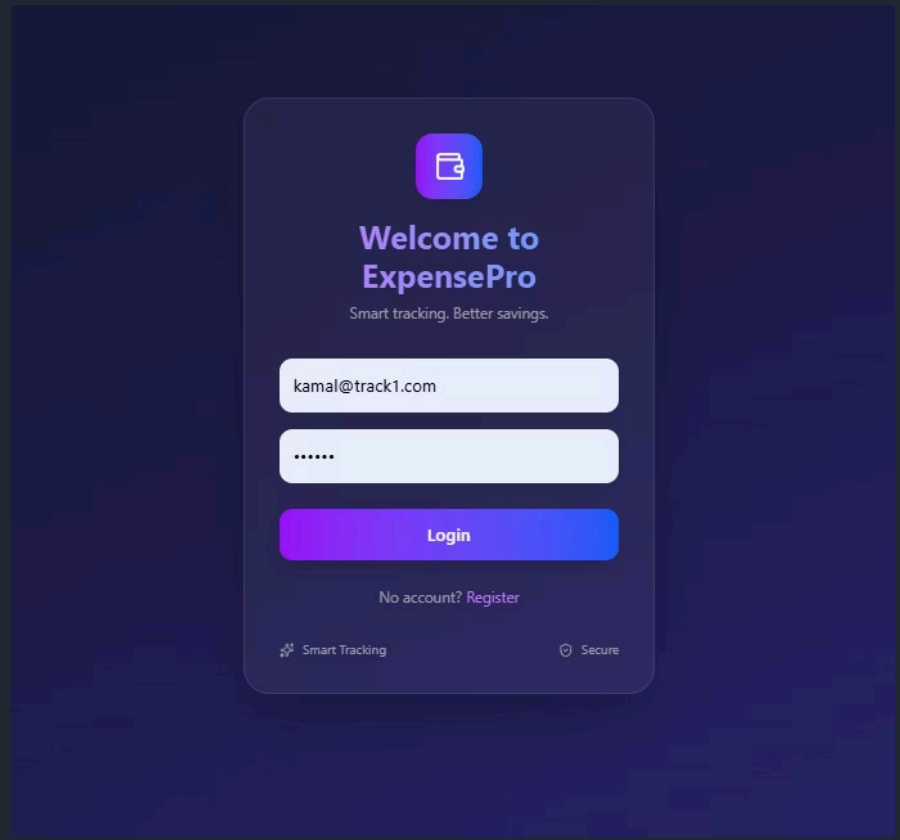
Category-Based Management

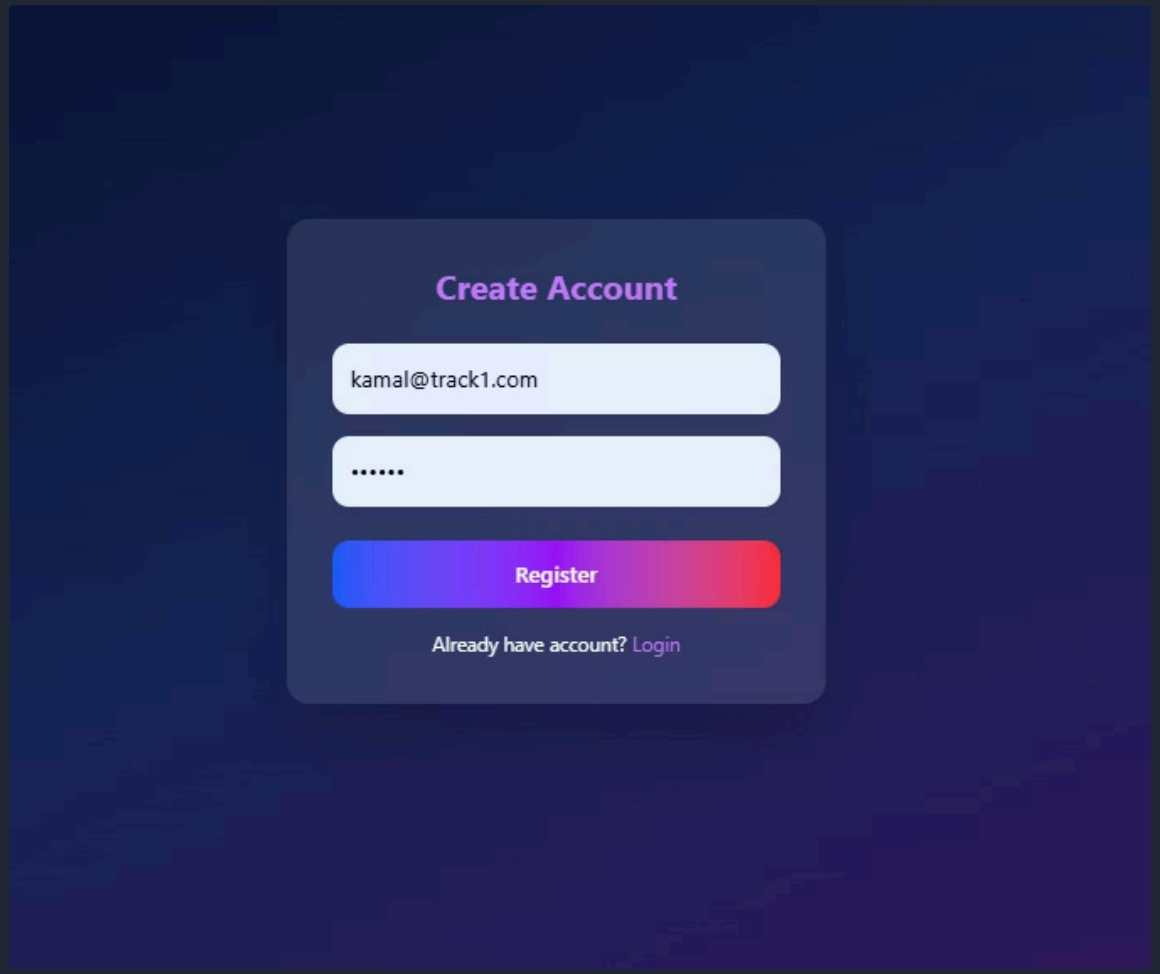
Expenses are categorized into groups such as: Food, Travel, Bills, Shopping, Entertainment, and Health.

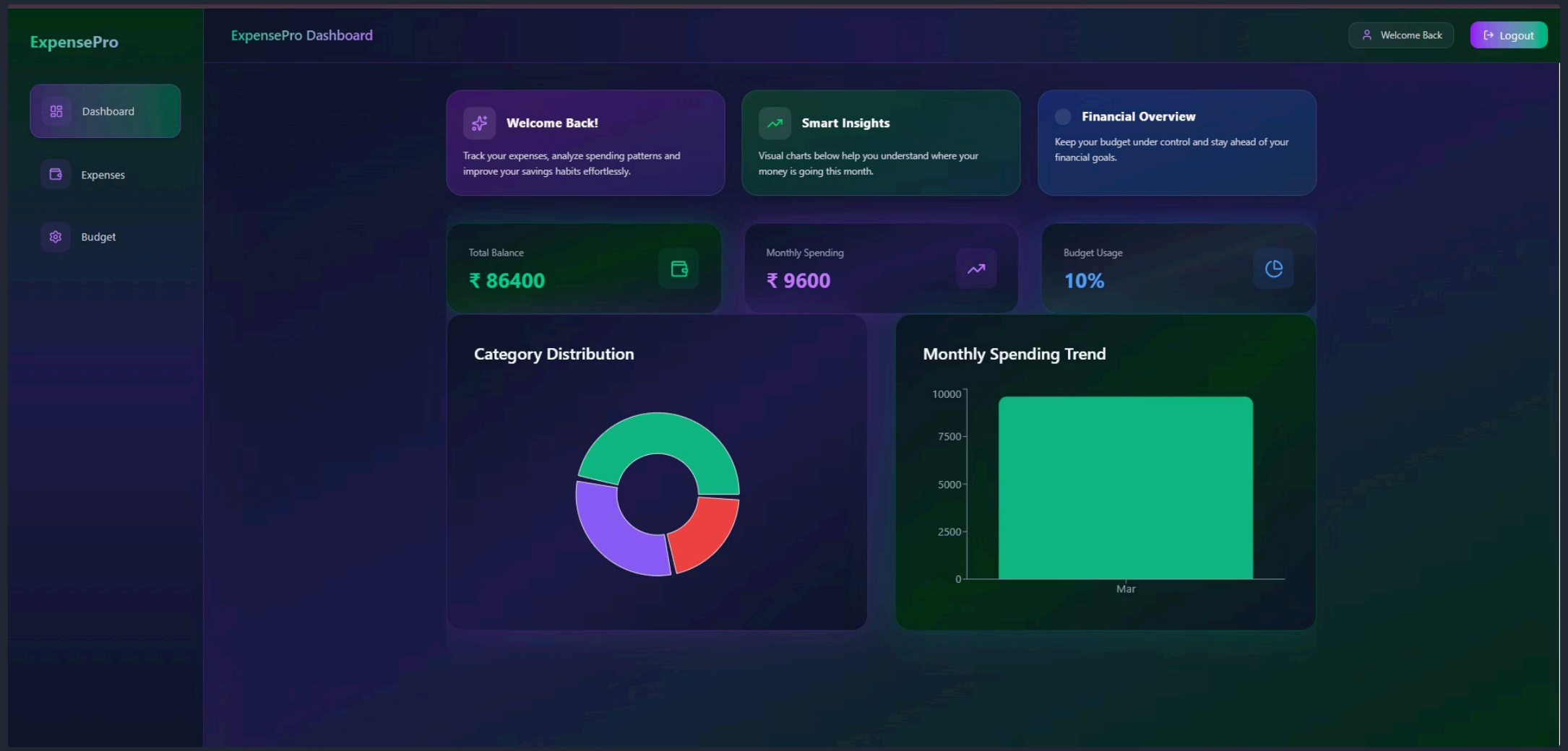
Dashboard Insights

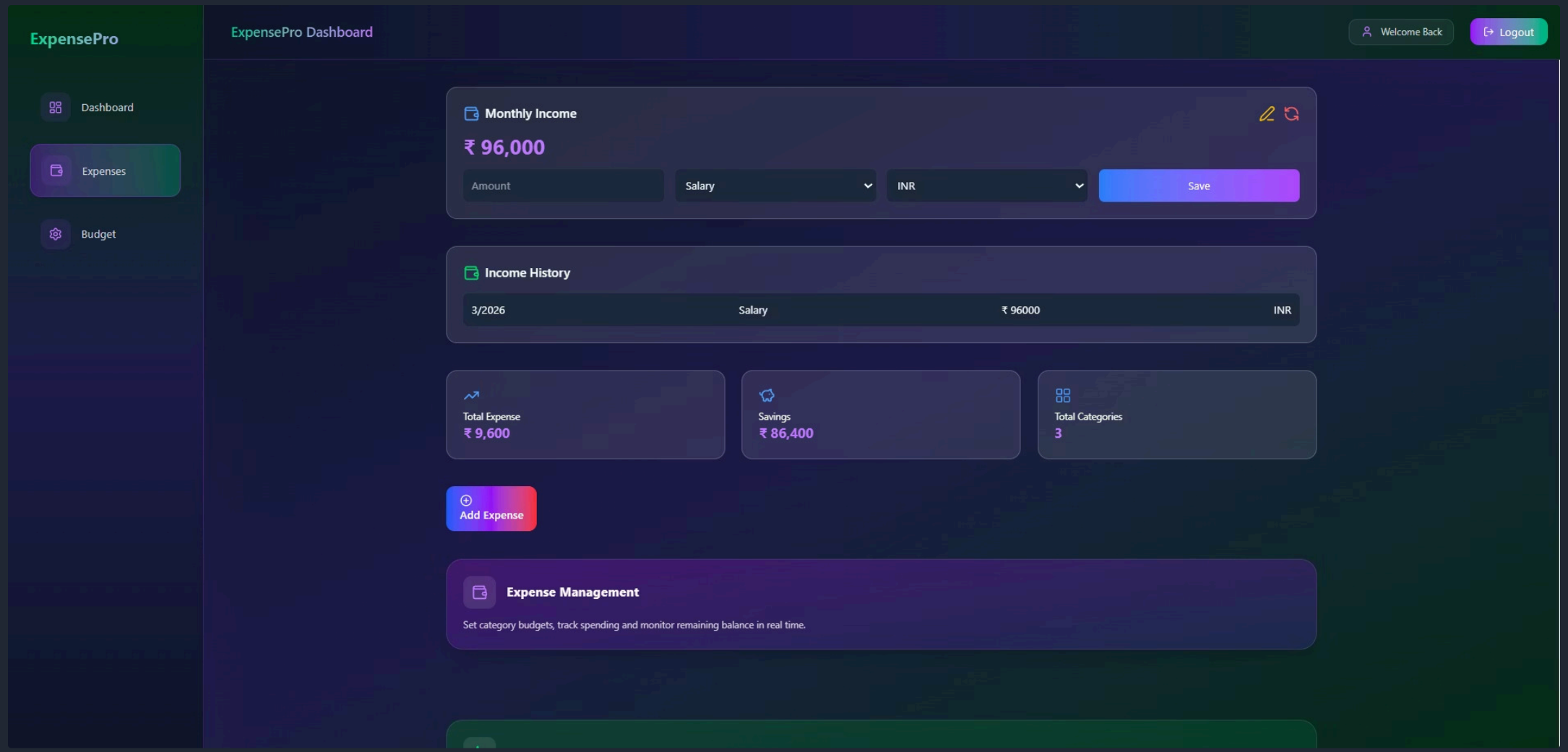
The dashboard provides financial insights and summaries.

ExpensePro Features Screenshots






ExpensePro
ExpensePro Dashboard
Welcome Back
Logout
Dashboard
Expenses
Budget
Welcome Back!
Track your expenses, analyze spending patterns and improve your savings habits effortlessly.
Smart Insights
Visual charts below help you understand where your money is going this month.
Financial Overview
Keep your budget under control and stay ahead of your financial goals.
Total Balance
₹ 86400
Monthly Spending
₹ 9600
Budget Usage
10%
Category Distribution
Monthly Spending Trend
10000
7500
5000
2500
0
Mar


ExpensePro
ExpensePro Dashboard
Welcome Back
Logout
Dashboard
Expenses
Budget
Monthly Income
₹ 96,000
Amount
Salary
INR
Save
Income History
3/2026
Salary
₹ 96000
INR
Total Expense
₹ 9,600
Savings
₹ 86,400
Total Categories
3
Add Expense
Expense Management
Set category budgets, track spending and monitor remaining balance in real time.

Live Tracking

Progress bars update automatically as you add expenses.

Category Budget Allocation

Food ₹2000 / ₹6001 6001 Remaining: ₹4001	**Travel** ₹3000 / ₹10000 10000 Remaining: ₹7000	**Bills** ₹4600 / ₹0 0 Remaining: ₹-4600
Shopping ₹0 / ₹0 0 Remaining: ₹0	**Health** ₹0 / ₹0 0 Remaining: ₹0	**Education** ₹0 / ₹0 0 Remaining: ₹0
Entertainment ₹0 / ₹0 0 Remaining: ₹0	**Rent** ₹0 / ₹0 0 Remaining: ₹0	**Insurance** ₹0 / ₹0 0 Remaining: ₹0

Expense Summary

March 2026

#	Amount	Category	Date	Action
1	600 INR	Bills	Tue Mar 03 2026	
2	3000 INR	Bills	Tue Mar 03 2026	
3	1000 INR	Bills	Tue Mar 03 2026	
4	3000 INR	Travel	Tue Mar 03 2026	
5	2000 INR	Food	Mon Mar 02 2026	

Budget Protection

Set your monthly budget and get alerts when spending approaches the limit.

Spending Warning

If your expenses exceed safe limits, you'll receive visual alerts.

Monthly Budget Setup

Set Monthly Budget

Save

Budget	Spent	Remaining
₹ 100,000	₹ 9,600	₹ 90,400

Usage 10%

Safe

Expense Management System

The expense management module allows users to monitor and control their spending.

Users can:

- Add new expenses
- Edit existing expenses
- Delete expense records
- Track expense categories

The system calculates:

- Total Expenses
- Remaining Balance
- Category Spending Distribution

These features help users maintain accurate financial records.

Page 9 — Budget Management

The budget management system allows users to set monthly spending limits.

1

Users can:

- Define a monthly budget
- Track spending against the budget
- Monitor remaining funds

2

The system automatically calculates:

- Budget usage percentage
- Remaining balance
- Spending alerts

This feature helps users control overspending and maintain financial discipline.

Page 10 — Conclusion

ExpensePro is a powerful and modern expense management system developed using the MERN stack. The platform enables users to manage their finances efficiently through intuitive interfaces and secure backend architecture.

The system includes features such as secure authentication, expense tracking, category management, financial insights, and budget monitoring.

By combining modern technologies with practical financial tools, ExpensePro provides an effective solution for personal expense management.

ExpensePro demonstrates the practical application of full-stack development using the MERN ecosystem.

In the future, the system can be enhanced with features such as:

- Mobile application support
- Advanced financial analytics
- Automated spending alerts
- Cloud deployment